

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2018

CL 317 Design of Structures-I

Date: 10.05.2018, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, SP 16, SP 6 and steel table are permitted, provided that they do not contain anything other than the printed matter inside.

SECTION – I

Q - 1 Answers the followings:

[07]

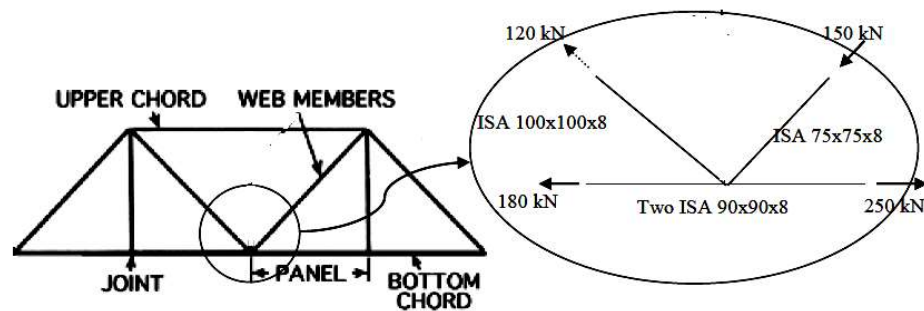
- (a) Modes of failure of tension members are
- | | |
|-----------------------------------|-----------------------------|
| (i) Rupture at a critical section | (ii) Yielding of gross area |
| (iii) Block shear of end region | (iv) All of these |
- (b) As compared to field bolts, the shop bolts are
- | | |
|----------------------|-------------------|
| (i) Stronger | (ii) Weaker |
| (iii) Equally strong | (iv) Any of these |
- (c) The heaviest I-section for same depth is
- | | |
|------------|-----------|
| (i) ISMB | (ii) ISLB |
| (iii) ISHB | (iv) ISWB |
- (d) Cross-sections in which the elements buckle locally even before reaching yield stress
- | | |
|----------------------------|----------------------|
| (i) Plastic section | (ii) Compact section |
| (iii) Semi compact section | (iv) Slender section |
- (e) Distance from the center of a fastener hole to the nearest edge of an element measured perpendicular to the direction of load transfer is known as
- | | |
|------------------|--------------------|
| (i) End distance | (ii) Edge distance |
| (iii) Pitch | (iv) Gauge |
- (f) The effective length of a column of length L fixed against rotation and translation at one end and free at the other end is
- | | |
|----------------|-------------|
| (i) $0.5L$ | (ii) $0.7L$ |
| (iii) $1.414L$ | (iv) $2L$ |
- (g) Net sectional area of a tension member, is equal to its gross section area _____.
- | | |
|--|--|
| (i) Plus the area of bolt holes | (ii) Divided by the area of bolt holes |
| (iii) Multiplied by the area of bolt holes | (iv) Minus the area of the bolt holes |

Q – 2(a) Explain the following terms used in bolted connections and show by means of sketches: **(Any TWO)**

- (i) Gauge distance (ii) Tacking bolts (iii) Pitch of the bolts (iv) Edge distance

Q - 2 (b) Attempt any TWO:**[12]**

- 1) One steel foot-over bridge at the Anand railway station failed due to sudden earthquake shock. While checking that bridge you have observed that one junction on the bridge where four angle sections were meeting was failing due to insufficient capacity of the bolt. You are now advised to redesign this junction for the safety of the people. The analysis results are as below in the **Figure 1**. You are also advised to use only M16 bolts of grade 4.6 as per availability in the market. Grade of plate 410. Also provide the drawing for proper implementation on site.

**Figure 1**

- 2) A single unequal angle $90 \times 60 \times 10$ mm is connected to a 12 mm thick gusset plate with 90 mm leg at the ends with 5 nos. of 18 mm diameter bolts to transfer tension. Determine the design tensile strength of the angle. Take Fe 410 grade steel, $f_y = 250$ MPa and 4.6 grade bolt. Take pitch of the bolt = 50 mm and end distance = 40 mm.
- 3) Design a simply supported beam over a span of 4 m which carries a factored dead load of 20 kN/m and factored live load of 20 kN/m, if the beam is fully restrained laterally. Check for web buckling and web crippling is not required. Take grade of steel plate as Fe410 and $f_y = 250$ MPa.

Q – 3(a) Explain following types of welded connections by means of sketches: (Any TWO) [02]

(i) Fillet weld (ii) Butt weld (iii) Slot weld (iv) Plug weld

Q – 3(b) Attempt any TWO:**[12]**

- 1) A tie plate of 100×8 mm is connected to the gusset plate to transmit a factored load of 120 kN. Determine the size and length of the fillet weld. Assume site welds and grade of steel plate as Fe410.
- 2) A 5-meter-long column is effectively held in position and restraint against rotation at both ends. If ISHB400 @ 74.5 kg/m is used, calculate the compressive load carrying capacity of the column. Take grade of steel plate as Fe410 and $f_y = 250$ MPa.

- 3) Design a simply supported beam over a span of 4 m which carries a factored dead load of 20 kN/m and factored live load of 20 kN/m, if the compression flange is laterally unsupported. Check for web buckling and web crippling is not required. Take grade of steel plate as Fe410 and $f_y = 250$ MPa.

SECTION – II

Q – 4 Answers the followings:

[07]

- (a) In an RCC Beam side face reinforcement is provided if its depth exceeds
- | | |
|--------------|-------------|
| (i) 650 mm | (ii) 750 mm |
| (iii) 700 mm | (iv) 600 mm |
- (b) The column bars of diameters larger than _____ in compression can be dowelled at the footings.
- | | |
|-------------|------------|
| (i) 36 mm | (ii) 32 mm |
| (iii) 24 mm | (iv) 30 mm |
- (c) The strength of compression members with helical reinforcement is _____ times of members with vertical ties?
- | | |
|------------|----------|
| (i) 1.5 | (ii) 1.2 |
| (iii) 1.05 | (iv) 1.1 |
- (d) A reinforced concrete slab is 80 mm thick the maximum size of reinforcement bar can be used is _____.
- | | |
|-------------|------------|
| (i) 8 mm | (ii) 10 mm |
| (iii) 12 mm | (iv) 16 mm |
- (e) What is the maximum grade of steel provided in stirrups _____?
- | | |
|-------------|------------|
| (i) Fe250 | (ii) Fe415 |
| (iii) Fe450 | (iv) Fe500 |
- (f) According to IS : 456 -2000, the modulus of elasticity of concrete E_c (in N/mm²) can be taken as _____
- | | |
|---------------------------|--------------------------|
| (i) $4500\sqrt{f_{ck}}$ | (ii) $5500\sqrt{f_{ck}}$ |
| (iii) $5000\sqrt{f_{ck}}$ | (iv) $6000\sqrt{f_{ck}}$ |
- (g) What is the spacing of shear reinforcement in slab if needs to be provided
- | | |
|------------|------------|
| (i) d | (ii) 1.1d |
| (iii) 0.5d | (iv) 0.75d |

Q – 5(a) Why is it necessary to have lateral ties in a column? What are the specifications for these laterals? **[02]**

(b) What is punching shear in RC footing? **[02]**

- (c) A beam, simply supported over an effective span of 8 m carries a live load of 20 kN/m. [10]
Design and detail this beam for flexure using M20 grade concrete and HYSD bars of grade Fe415. Assume unit weight of concrete as 25 kN/m³. Check this beam for deflection and cracking only.

OR

- Q – 5(a)** A RC beam 300 mm × 700 mm in cross-section, is reinforced with 4 Nos of 25 mm diameter & 2 Nos of 20 mm diameter in tension bars of grade Fe415, with an effective cover of 70 mm. The ultimate shear at the section is 242 kN. Design and detail the shear reinforcement. [06]
- (b) A solid footing has to transfer a load of 600 kN from a square column 230 mm × 230 mm with 16 mm diameter bars. Design and detail the isolated uniform depth footing. Soil bearing capacity of soil is 200 kN/m². Use M20 and Fe415. Do check for two way shear only. [08]
- Q-6(a)** Why torsional reinforcements are required in two way slab for corners lifted down. [02]
- (b) Design and detail a simply supported R.C.C slab for a roof of a hall 4 m × 10 m (inside dimensions) with 230 mm walls all around. Assume a live load of 4 kN/m² and floor finish 1 kN/m². Use M25 and Fe415. [05]
- (c) Design and detail the short column for the following data if the moment due to minimum eccentricity is less than the applied moment. [07]
Column size: 400 mm × 400 mm, $P_u = 960$ kN, $M_{ux} = 300$ kNm. Use M20 and Fe415.

OR

- Q-6(a)** Design and detail a simply supported square two-way slab supported on 250 mm wide supports at a distance of 4.5 m centre-centre. The slab is subjected to a live load of 4 kN/m² and a surface finish of 1 kN/m². Consider that the corners are held up. Consider M25 and Fe500. [07]
- (b) What is the maximum percentage of steel allowed in RC column? Explain why it is necessary to specify this maximum percentage. [07]
Design the reinforcement in a column of size 400 mm x 600 mm subjected to an axial load of 2000 kN under service dead load and live load. The column has an unsupported length of 4.0 m and effectively held in position and restrained against rotation in both ends. Use M25 concrete and Fe415 steel.
